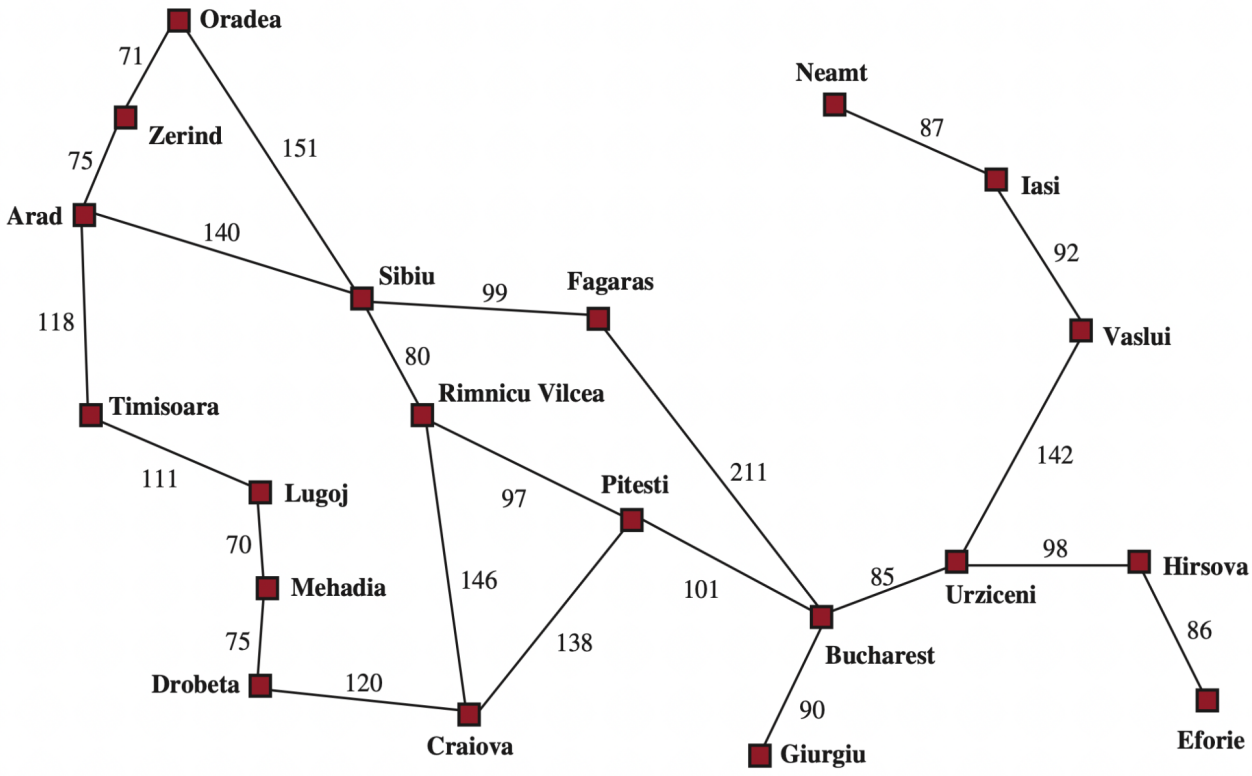
AL-2002 - Artificial Intelligence Lab
Lab # 5

Objectives:

- Hill climbing + Local Beam Search + Simulated Annealing

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your program.
2. Write Program in multiple cells of the same notebook**.**
3. **Must Use Functions and Exception handling for each question, where they can be applied.**
4. Do not copy from any source otherwise you will be penalized with negative marks.
5. Add your source code in word document and attach output screenshots.
6. Upload word file and ipynb file.
7. Please submit your **Both files** with this naming convention ROLLNO_SECTION_LABNO. **(Do not upload zip file).**
8. Submit your lab on Google Classroom.



Heuristic table: Euclidean distance to Bucharest (goal):

Arad 366	Bucharest 0
Craiova 160	Dobreta 242
Eforie 161	Fagaras 178
Giurgiu 77	Hirsova 151
Iasi 226	Lugoj 244
Mehadia 241	Neamt 234
Oradea 380	Pitesti 98
Rimnicu_Vilcea 193	Sibiu 253
Timisoara 329	Urziceni 80
Vaslui 199	Zerind 374

Question 1:

Using the provided weighted Romania road graph and your heuristic table of Euclidean distance to Bucharest (goal), run **hill climbing algorithm** from **Arad** to try to reach **Bucharest**: at each step, look only at the current city's immediate neighbors and move to the neighbor with the smallest heuristic value **h(n)** (if there is a tie, break it alphabetically); if no neighbor has a strictly smaller **h** than the current city, stop and report that you are stuck in a local minimum/plateau. Write the full sequence of visited cities (showing each city's h value), then compute the total path cost by summing the road distances along the edges you traverse.

Start: **Arad**
Goal: **Bucharest**

Question 2:

Using the provided **weighted Romania road graph** and the **heuristic table of Euclidean distance to Bucharest (goal)**, run the **Local Beam Search algorithm** with **beam width** $B = 2$ to try to reach **Bucharest**.

At each step:

1. Start with the **initial beam containing the start city Arad**.
2. Generate **all immediate neighbors** of every city currently in the beam.
3. From all generated neighbors, **select the two cities with the smallest heuristic values** $h(n)$ to form the **new beam**.
4. If there is a **tie**, break it **alphabetically**.
5. If **Bucharest is selected**, stop the search.
6. If **no newly generated neighbor has a strictly smaller heuristic value than the best city in the current beam**, stop and report that the search is **stuck in a local minimum or plateau**.

Tasks:

1. Write the **beam at each step**, showing the cities and their heuristic values $h(n)$.
2. Write the **final sequence of cities visited that leads to Bucharest (if reached)**.
3. Compute the **total path cost** by summing the road distances along the edges traversed.

Start: Arad
Goal: Bucharest
Beam width: $B = 2$

Question 3:

```
SIMULATED_ANNEALING(initial_state):

    current ← initial_state
    best ← current
    T ← initial_temperature

    while T > stopping_temperature:

        next ← RANDOM_NEIGHBOR(current)

        ΔE ← COST(next) − COST(current)

        if ΔE < 0 then
            current ← next          // better solution
        else
            p ← random number between 0 and 1
            if p < exp(−ΔE / T) then
                current ← next      // accept worse solution with probability
            end if
        end if

        if COST(current) < COST(best) then
            best ← current
        end if

        T ← COOL(T)                 // reduce temperature

    return best
```

Using the provided **weighted Romania road graph** and the **heuristic table of Euclidean distance to Bucharest (goal)**, run the **Simulated Annealing algorithm** to try to reach **Bucharest** starting from **Arad**.

At each step:

1. Start with the **initial state as the start city Arad**.
2. Set the **initial temperature** $T = 100$.
3. Select **one immediate neighbor** of the current city (consider neighbors in **alphabetical order**).
4. Compute the **change in heuristic value**

$$\Delta h = h(\text{neighbor}) - h(\text{current})$$

5. If the neighbor has a **smaller heuristic value**, move to that city.
6. If the neighbor has a **larger heuristic value**, accept the move with probability

$$P = e^{-\Delta h/T}$$

7. Reduce the temperature using the **cooling schedule**

$$T = 0.8 \times T$$

8. If **Bucharest is reached**, stop the search.

9. If the **temperature falls below** $T < 1$, stop and report that the search has terminated due to **low temperature**.

Tasks:

1. Write the **sequence of cities visited**, showing the **heuristic value** $h(n)$ and the **temperature** T at each step.
2. Indicate whether each move was **accepted because it improved the heuristic** or **accepted probabilistically using the simulated annealing rule**.
3. Compute the **acceptance probability** P whenever a worse move is considered.
4. Compute the **total path cost** by summing the **road distances along the edges traversed**.

Start: Arad
Goal: Bucharest
Initial Temperature: $T = 100$
Cooling Schedule: $T = 0.8 \times T$

Hint:

```
import math

delta_h = 76
T = 100
P = math.exp(-delta_h / T)
```